HW 2

Team members: Catherine Chen, Dylan Pan, Hsiao-Chi Chen, Jackie Zhang, Zara Nip

Financial Cycles

Q1. The market pricing to occur to the Federal Funds rate by end of next year

We infer the expected policy rate using the December 2027 Fed Funds futures contract (CME ticker: ZQZ7), which is quoted as 100 minus the implied policy rate. As of April 24, 2026, the contract trades at 96.585, implying a Fed Funds rate of 3.415% by end-2027.

The current FOMC target range is 3.50%–3.75% (midpoint: 3.625%), so this implies about 21 bps of easing, or 0.84 cuts of 25 bps.

The CME FedWatch tool, which derives rate probabilities from Fed Fund futures, supports this pricing for the December 2027 meeting. Most probability mass lies in the 300–350 bps range, below the current policy range, while only 10.2% remaining in the current 350–375 bps range.

Overall, the market is pricing a modest easing path, with less than one rate cut expected by the end of next year.

Table: FedWatch- Target Rate Probability

	CME FEDWATCH TOOL - CONDITIONAL MEETING PROBABILITIES							
MEETING DATE	225-250	250-275	275-300	300-325	325-350	350-375	375-400	400-425
2026/4/29	0.0%	0.0%	0.0%	0.0%	0.0%	100.0%	0.0%	0.0%
2026/6/17	0.0%	0.0%	0.0%	0.0%	3.2%	96.8%	0.0%	0.0%
2026/7/29	0.0%	0.0%	0.0%	0.1%	7.2%	92.6%	0.0%	0.0%
2026/9/16	0.0%	0.0%	0.0%	0.4%	10.8%	88.7%	0.0%	0.0%
2026/10/28	0.0%	0.0%	0.0%	0.4%	10.6%	87.0%	2.0%	0.0%
2026/12/9	0.0%	0.0%	0.0%	0.4%	10.6%	87.0%	2.0%	0.0%
2027/1/27	0.0%	0.0%	0.0%	0.4%	9.8%	80.8%	8.8%	0.2%
2027/3/17	0.0%	0.0%	0.0%	0.4%	9.0%	74.8%	14.9%	0.9%
2027/4/28	0.0%	0.0%	0.0%	0.3%	8.4%	69.9%	19.4%	1.9%
2027/6/9	0.0%	0.0%	0.0%	0.9%	12.4%	66.6%	18.3%	1.8%
2027/7/28	0.0%	0.0%	0.1%	1.4%	14.8%	64.4%	17.5%	1.7%
2027/9/15	0.0%	0.0%	0.3%	4.0%	24.4%	55.4%	14.5%	1.4%
2027/10/27	0.0%	0.1%	1.4%	9.8%	33.3%	43.6%	10.7%	1.0%
2027/12/8	0.2%	1.8%	11.0%	33.8%	41.9%	10.2%	1.0%	0.0%

Q2 Historical data suggestion in cutting rate cycle

We define a cutting period as a >= 50 bps decline in the Fed Funds rate over the six months. Since 1990, we compare SPY, 10Y Treasury total return estimated from FRED DGS10 yields, DBC, and HYG using annualized return, volatility, and positive-month frequency.

During cutting periods, bonds performed best, earning 6.37% annualized with 8.40% volatility and a 62.1% positive-month frequency. Risk assets struggled: equities returned -6.62%, commodities -14.72%, and high-yield credit -3.31% annualized.

Table: Asset Performance in Fed Cutting vs Non-Cutting Regimes (Since 1990)

	Cutting Ann. Ret %	Cutting Ann. Vol %	Cutting % months > 0	Non-Cutting Ann. Ret %	Non-Cutting Ann. Vol %	Non-Cutting % months > 0
US Equity	-6.62	20.19	31.6	15.46	13.36	65.2
US Bond	6.37	8.40	62.1	4.65	7.10	57.2
Commodities	-14.72	26.40	17.9	5.36	16.44	28.9
US HY Credit	-3.31	18.99	14.7	6.85	7.75	36.3

Moreover, looking at 12-month forward returns after easing cycles, bonds are the most consistent performer, with positive returns in 6 of 7 episodes and an average return of 5.9%. Equities have higher average returns of 10.9% but greater dispersion, performing poorly in recessions but well in soft landings. Commodities and high-yield credit are less reliable.

Overall, bonds are the most reliable beneficiary of easing, while equities offer conditional upside depending on the macro environment.

Table: 12-month forward returns at each cutting cycle

	US Equity	US Bond	Commodities	US HY Credit
1990-07	NaN	9.80	NaN	NaN
1995-07	25.99	2.10	NaN	NaN
1998-09	39.27	-2.38	NaN	NaN
2001-01	-11.75	5.05	NaN	NaN
2007-09	-10.99	9.84	51.48	-1.22
2019-08	12.00	15.62	-15.45	3.33
2024-09	NaN	1.31	NaN	NaN

Q3. Does This Extrapolation Make Sense for Now?

This extrapolation makes sense only partly. Fed cuts historically support long-

duration bonds by lowering yields, but today's setup is not an aggressive easing cycle. Q1 shows the market prices only about 0.84 cuts by end-2027, leaving limited easing upside unless growth weakens or the Fed turns more dovish.

Equity extrapolation is more conditional. Stocks can benefit in soft landings, but recessionary cuts often bring earnings pressure and valuation compression. With equities already expensive, historical average returns should not be treated as unconditional.

Commodities and high-yield credit are weaker easing trades. In our cutting-regime sample, both delivered negative annualized returns and low positive-month frequencies, implying greater dependence on growth and risk appetite than on rate cuts alone.

Thus, long duration is the cleanest Fed easing expression. Other risk assets require a constructive growth view, so the historical cutting-cycle playbook should be applied cautiously.

Macro regimes

Q4. Two Macro Variables That Matter for Asset Prices

We use CPI year-over-year inflation (CPIAUCSL) and industrial production year-over-year growth (INDPRO). CPI measures inflation pressure, while industrial production measures real activity. Both are monthly series available before 1995.

Inflation matters through policy expectations and discount rates: higher inflation can raise yields, hurt bonds and equity valuations, and support commodities.

Growth matters through earnings, credit quality, and risk appetite: stronger growth supports equities and high-yield credit, while weaker growth favors government bonds through lower-rate expectations and flight-to-quality demand.

Together, these two variables create a simple 2×2 macro regime framework: growth up/down × inflation up/down.

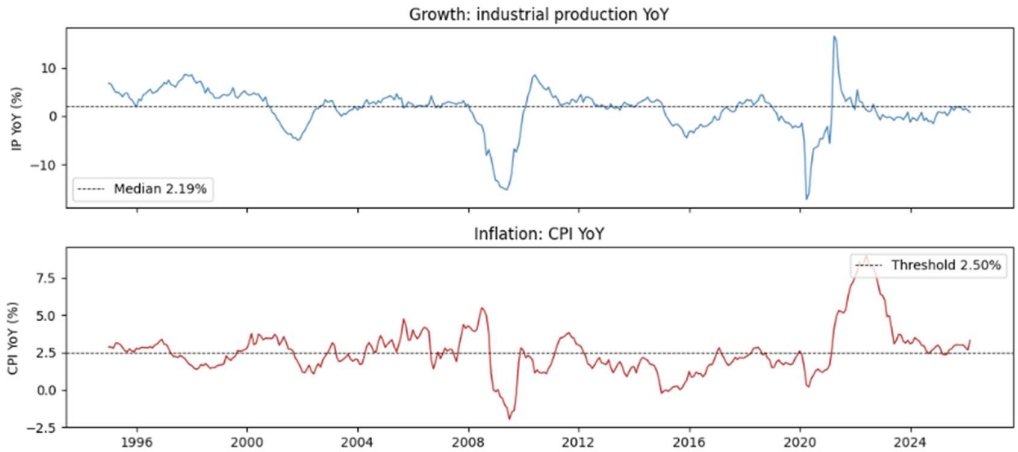
Table: Macro Regime Framework

	Inflation Down	Inflation Up
Growth Up	Goldilocks / Disinflationary Growth	Reflation
Growth Down	Deflationary Slowdown	Stagflation

Q5. Defining Macro Regimes Since 1995

We define regimes using the two variables from Q4: industrial production YoY growth and CPI YoY inflation. The first chart shows the thresholds used for classification: growth is split by the post-1995 median IP YoY growth rate of 2.19%, while inflation is split using a 2.50% CPI YoY threshold, which is close to the Fed's inflation target with a small buffer.

Chart: Growth and Inflation Thresholds for Macro Regime Classification



The regime-spell table then shows the actual historical periods classified under each regime, requiring each spell to last at least six months. This avoids over-interpreting short month-to-month noise and highlights persistent macro environments.

Several key episodes line up well with economic history:

- Goldilocks: The clearest examples are the late-1990s expansion and 2010–2011 post-GFC recovery, when growth was solid while inflation stayed contained.
- Reflation/overheating: This regime captures periods such as 1995–1997, 1999–2000, 2004–2005, and 2021, when growth was strong but inflation pressure was also rising.
- Stagflation: The main episodes are the 2008 oil-price shock and the 2022–2024

inflationary slowdown, when growth weakened while inflation remained elevated.

- Disinflationary slowdown/recession: This regime lines up with the 2001 dot-com recession, the 2008–2009 GFC, the 2015–2016 manufacturing slump, and the 2020 COVID recession/recovery period.

Table: Macro Regime Spells Since 1995 Based on Growth and Inflation Conditions

```
Regime spells of at least 6 months since 1995:
       regime      start        end  months
GrowUp/InflUp 1995-01-01 1995-12-01      12
GrowUp/InflUp 1996-02-01 1997-03-01      14
GrowUp/InflDn 1997-04-01 1999-08-01      29
GrowUp/InflUp 1999-09-01 2000-10-01      14
GrowDn/InflUp 2000-11-01 2001-09-01      11
GrowDn/InflDn 2001-10-01 2002-09-01      12
GrowDn/InflDn 2003-04-01 2004-03-01      12
GrowUp/InflUp 2004-05-01 2005-08-01      16
GrowDn/InflUp 2008-02-01 2008-10-01       9
GrowDn/InflDn 2008-11-01 2009-11-01      13
GrowUp/InflDn 2010-03-01 2011-02-01      12
GrowUp/InflUp 2011-03-01 2012-03-01      13
GrowUp/InflDn 2012-04-01 2012-09-01       6
GrowUp/InflDn 2014-03-01 2015-01-01      11
GrowDn/InflDn 2015-02-01 2016-12-01      23
GrowDn/InflDn 2017-03-01 2017-09-01       7
GrowUp/InflDn 2017-10-01 2018-04-01       7
GrowDn/InflDn 2019-01-01 2019-12-01      12
GrowDn/InflDn 2020-02-01 2021-02-01      13
GrowUp/InflUp 2021-04-01 2021-12-01       9
GrowDn/InflUp 2022-10-01 2024-08-01      23
GrowDn/InflUp 2025-06-01 2026-03-01      10
```

Q6. Asset Performance Across Macro Regimes

The regime results show that growth is the key driver for bonds, while inflation is more important for HY credit, either can be a driver for equity, and only two together drive commodities.

Table: Macro Asset Performance Across Growth–Inflation Regimes Since 1995

	Ann Mean %				Ann Vol %				Sharpe			
Asset	Commodities	US Bond	US Equity	US HY Credit	Commodities	US Bond	US Equity	US HY Credit	Commodities	US Bond	US Equity	US HY Credit
regime												
GrowDn/InflDn	1.82	3.48	13.83	10.41	19.09	8.34	15.91	12.18	0.10	0.42	0.87	0.85
GrowDn/InflUp	-6.38	1.40	0.07	-0.97	19.46	7.63	17.14	11.67	-0.33	0.18	0.00	-0.08
GrowUp/InflDn	-2.86	5.91	16.18	5.31	16.63	6.87	14.99	7.05	-0.17	0.86	1.08	0.75
GrowUp/InflUp	18.60	6.51	12.44	-0.17	17.78	6.93	12.64	8.53	1.05	0.94	0.98	-0.02

Equities are broadly resilient except in Stagflation. Equities perform best when either

growth-up or inflation-down is present, with a Sharpe ratio 0.87 in Deflationary Slowdonw, 1.08 in Goldilocks, and 0.98 in Reflation. In Stagflation, Sharpe ratio is nearly zero, meaning no excess returns, suggesting that the combination of weak growth and rising inflation is the most damaging environment for equities.

Government bonds are positive across all regimes, but their performance is weakest in Stagflation, with a 0.18 Sharpe ratio. They perform better in growth-up regimes (Reflation and Goldilocks), where Sharpe ratios are greater than 0.8.

Commodities are the clearest Reflation asset, earning a 1.05 Sharpe ratio when both growth and inflation are rising. Their losses in Goldilocks and Stagflation suggest that commodities benefit most when strong demand and rising inflation occur together.

High-yield credit performs best when inflation is falling, with a Sharpe ratio 0.85 in Deflationary Slowdown and 0.75 in Goldilocks. It weakens in both inflation-up regimes (Stagflation and Reflation), reflecting sensitivity to inflation pressure and credit-spread conditions.

Overall, Stagflation is the weakest regime, Reflation most clearly favors commodities, and Goldilocks provides the strongest environment for equities and broader risk assets.

Q7. Should We Use the Same Risk Metrics Across Regimes?

We should not use the same metrics across regimes. Volatility and Sharpe are useful baselines, but they assume returns are relatively symmetric and stable. This is more reasonable in Goldilocks and Reflation, where diversification tends to work better.

In stressed regimes, tail and correlation risks matter more. In Stagflation, we would focus on stock-bond correlation and inflation beta, since inflation shocks can hurt both stocks and bonds. In Deflationary Slowdown / Recession, we would emphasize maximum drawdown, CVaR / expected shortfall, and liquidity-adjusted volatility, especially for high-yield credit, where realized volatility can understate tail risk.

Across regimes, we would also monitor vol-of-vol, VIX, and MOVE, since volatility instability signals that normal risk assumptions are breaking down.

Overall, volatility and Sharpe are only the starting point. When macro conditions

deteriorate, drawdown, tail loss, correlation breakdown, and inflation sensitivity become more important.

Mini-project question: Risk Parity

Q8: Observing correlations

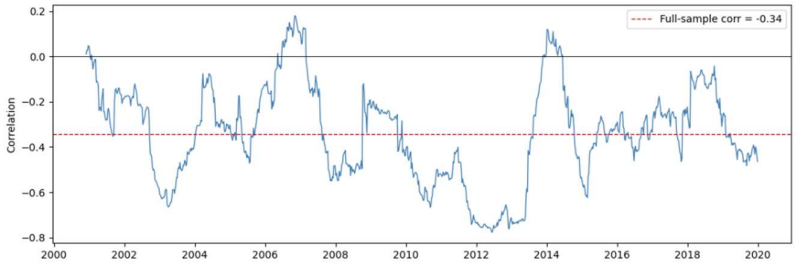
a) Full-Sample Correlation

The full-sample weekly equity-bond correlation from 2000 to 2019 is -0.34, meaning stocks and bonds generally move in opposite directions.

b) Rolling Correlation

The rolling 1-year correlation shows this relationship is not stable over time. While it is negative about 93% of the time, it varies widely, ranging from -0.78 to +0.~~/~~18, and occasionally turns positive.

Chart: Rolling 1-year equity-bond return weekly correlation



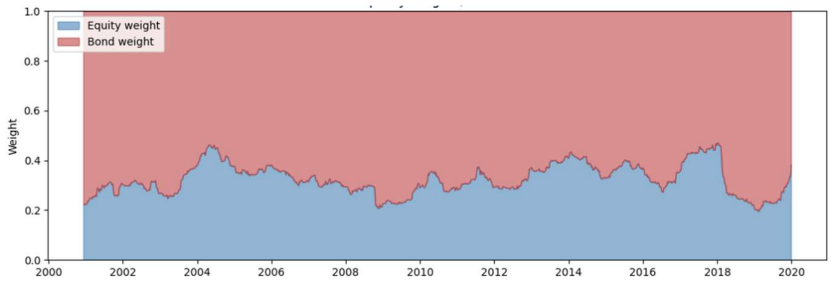
In conclusion, stocks and bonds usually provide diversification, but not always. The negative correlation is strongest during stress periods (flight to quality) and weaker or even positive when rates and inflation drive markets.

Q9: Historical Performance

a) Portfolio Allocation

We use a risk parity approach, allocating weights based on 1-year trailing volatility so that equities and bonds contribute equally to portfolio risk. On average, the portfolio allocates about 32.6% to equities and 67.4% to bonds. The portfolio is bond-heavy because bonds are less volatile, so a larger weight is needed to match equity risk. Over time, equity weights range from about 20% to 47%, rising in calm periods and falling when equity volatility increases.

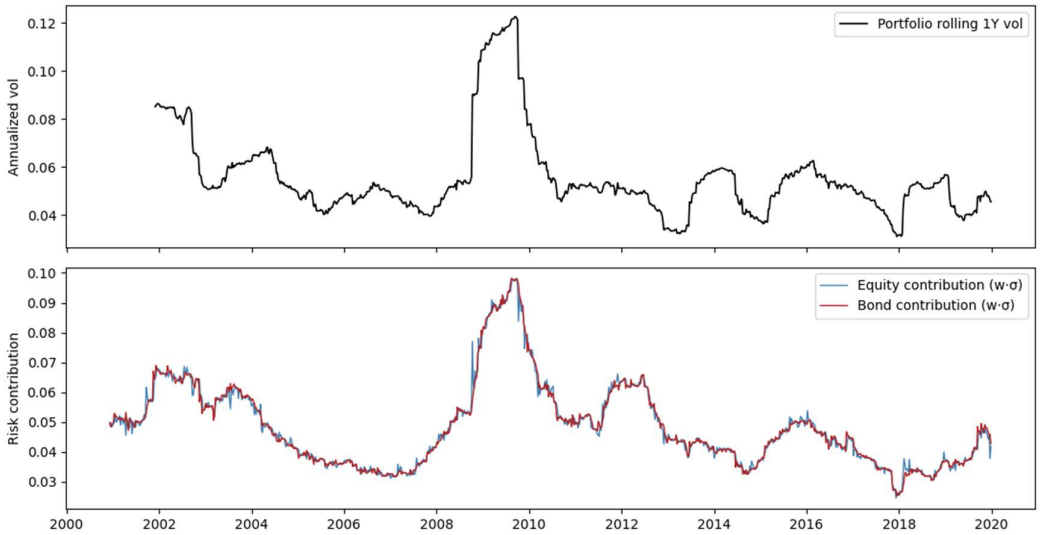
Chart: Risk-Parity Weights, 2000-2019



b) Portfolio Risk

The fund's average annualized risk from 2000–2019 is about 5.55%, with realized volatility usually around 3%–6%, but rising sharply to about 12% during the 2008–09 crisis. By construction, the strategy gives roughly equal standalone risk to equities and bonds: the average equity risk contribution is 4.85%, and the average bond risk contribution is 4.86%. The portfolio's total volatility is lower than the sum of these contributions because stocks and bonds were often negatively correlated, providing diversification.

Chart: Risk-Parity Fund Risk Over Time



c) Reasons for Risk Parity

Risk parity works well for stocks and bonds because of three key features. First, both asset classes have historically delivered similar Sharpe ratios, so equalizing risk can achieve strong risk-adjusted returns without forecasts. Second, stock–bond correlation has often been negative, which lowers overall portfolio risk through diversification. Third, bond volatility is much lower than equity volatility, meaning traditional portfolios are heavily exposed to equity risk. By allocating risk instead of capital, risk parity achieves better diversification and more balanced exposures. In

this dataset, it delivers about 6% return at 6% volatility, implying a Sharpe ratio close to 1 without leverage.

Q10: 2021 to the Present

a) Recent Year Performance

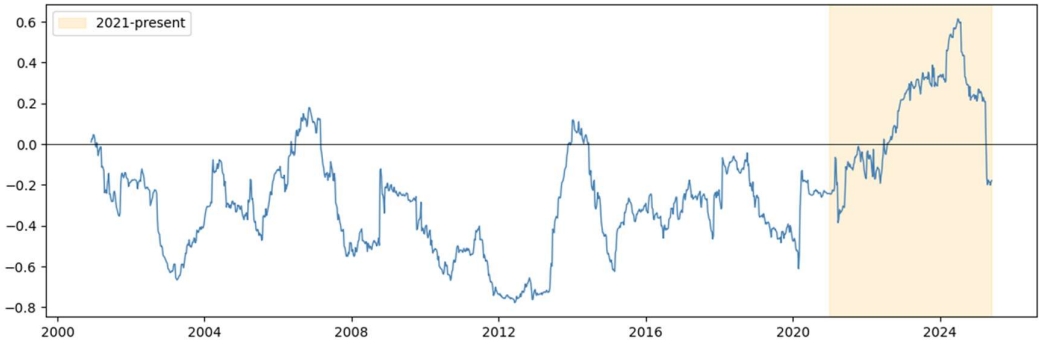
The strategy experienced highly uneven performance over this period. It suffered a large drawdown in 2022, with a -16.82% return. This was followed by a recovery in 2023 (+10.86%) and 2024 (+8.38%), and modest gains in 2025 so far. However, the cumulative return from 2021 to May 2025 is only ~4.2% with 8.24% annualized volatility, indicating weaker overall performance compared to the 2000–2019 period.

b) Why

The key reason is the breakdown of stock–bond diversification. The equity–bond correlation shifted from -0.34 (2000–2019) to +0.12 (2021–present), and the rolling correlation was frequently positive, peaking above 0.6.

Risk parity relies on bonds offsetting equity losses, but in 2022 both assets declined together due to high inflation and aggressive Fed tightening. With positive correlation, the portfolio lost its diversification benefit and became exposed to the same macro shock (rising rates), leading to simultaneous losses in both equities and bonds.

Chart: Rolling 1-year equity-bond correlation, 2000 - Present



c) Implications

Risk parity should not assume stock–bond correlation is always negative. When inflation drives markets, correlations can turn positive and undermine diversification. This suggests the need to adapt risk budgets to regimes, include inflation-sensitive assets (e.g., commodities or TIPS), and use more responsive risk

controls to reduce exposure when diversification breaks down.